

Homework (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x$, what is $(f + g)(x)$?
(A) $x^2 + 2x$
(B) $x^2 - 2x$
(C) $2x^2$
(D) $x^2 + x$
(E) $2x + x^2$
- Q2.** What is the domain of $(f/g)(x)$ if $f(x) = x + 1$ and $g(x) = x - 2$?
(A) $(-\infty, 2) \cup (2, \infty)$
(B) $(-\infty, 2)$
(C) $(2, \infty)$
(D) $[2, \infty)$
(E) $(-\infty, 2]$
- Q3.** A patient's dosage of medicine is given by $f(x) = 2x$ mg and $g(x) = x^2$ mg, where x is the number of hours since the first dose. What is the total dosage after x hours?
(A) $2x + x^2$ mg
(B) $2x - x^2$ mg
(C) $x^2 - 2x$ mg
(D) $2x^2$ mg
(E) x^2 mg
- Q4.** If $f(x) = x^2 - 1$ and $g(x) = x + 2$, what is $(f \cdot g)(x)$?
(A) $x^3 + x^2 - 2x - 2$
(B) $x^3 + x^2 - x - 2$
(C) $x^2 + 2x - 1$
(D) $x^2 - 2x - 2$
(E) $x^3 - x^2 - 2x + 2$
- Q5.** What is the result of $(f/g)(x)$ if $f(x) = x^2 + 1$ and $g(x) = x - 1$?
(A) $\frac{x^2+1}{x-1}$
(B) $\frac{x+1}{x-1}$
(C) $x + 1$
(D) $x - 1$
(E) $\frac{x^2-1}{x+1}$
- Q6.** The growth rate of a cell culture is given by $f(x) = 2x$ cells/hour and $g(x) = x^2$ cells/hour, where x is the number of hours. What is the combined growth rate after x hours?
(A) $2x + x^2$ cells/hour
(B) $2x - x^2$ cells/hour
(C) $x^2 - 2x$ cells/hour
(D) $2x^2$ cells/hour
(E) x^2 cells/hour
- Q7.** If $f(x) = x + 2$ and $g(x) = x - 3$, what is $(f - g)(x)$?
(A) $x + 2 - (x - 3)$
(B) $x + 2 + x - 3$
(C) $x - 3 - (x + 2)$
(D) $x + 2 + 3$
(E) 5
- Q8.** The probability of a patient's recovery is given by $f(x) = x^2$ and $g(x) = 1 - x$, where x is the number of treatments. What is the combined probability after x treatments?
(A) $x^2 + (1 - x)$
(B) $x^2 - (1 - x)$
(C) $x^2(1 - x)$
(D) $x^2 + x$
(E) $1 - x^2$

Open Ended Questions (FRQ)

- Q9.** A doctor prescribes a patient with two types of medicine. The dosage of the first medicine is given by $f(x) = x + 1$ mg, where x is the number of hours since the first dose. The dosage of the second medicine is given by $g(x) = 2x$ mg. Find the total dosage after x hours and simplify the resulting expression.
- Q10.** In a medical study, the growth rate of a cell culture is modeled by two functions: $f(x) = x^2 + 1$ cells/hour and $g(x) = x - 1$ cells/hour. Find the combined growth rate after x hours and determine the domain of the combined function.

Chef's Tips

- Review the learning objectives for Unit 3: Functions
- Emphasize the importance of simplifying resulting expressions
- Provide examples of real-world applications of function operations in healthcare and biology